

GIANLUIGI VITALE

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EDUCATION

B.Sc. in Computer Engineering — *Universitas Mercatorum, Rome, Italy* *Expected Feb. 2027*

GPA: 29.03/30 (Italian scale) · 3.79/4.0 (WES-converted)

Thesis: DSProfiler - Static Analysis and Machine Learning for Automated Detection of Data Structure

Performance Anti-patterns in Python

Supervisor: Prof. Rocco Pietrini

Scientific High School Diploma — *Liceo Scientifico "S. Cannizzaro," Palermo* *2014–2019*

Full Stack Web Development Bootcamp — *Boolean SRL (700 hours)* *Jan.–Jun. 2020*

PUBLICATIONS

G. Vitale, "DriftBench: Measuring and Predicting Infrastructure Drift in LLM Serving Systems," *Proceedings of MLSys 2026*, Bellevue, WA, May 2026. Oral presentation. DOI: 10.5281/zenodo.19361066

UNDER REVIEW

G. Vitale, "Decomposing One Professional-Framing Pipeline: Which Components Shift LLM Safety Boundaries?," *Submitted to NeurIPS 2026 main track*.

G. Vitale, "Natural Language Is Flatland for Machine Intelligence: AI Systems Should Decouple Their Reasoning Substrate from Human Language," Submitted to NeurIPS 2026 position paper track.

RESEARCH INTERESTS

Machine learning systems, LLM serving infrastructure, ML benchmarking, inference reliability, LLM safety evaluation, prompt robustness, retrieval-augmented generation security.

PROFESSIONAL EXPERIENCE

Systems Engineer — *Italian Ministry of Culture* *Apr. 2024–Present*

Designed and deployed a Retrieval-Augmented Generation system over 10,000 internal administrative documents, delivering source-cited answers to staff queries while minimizing hallucinations. Pioneered the first AI-assisted public consultation interface in an Italian state archive, enabling both employees and citizens to search and cross-reference the full inventory database through natural language, fundamentally transforming how the institution serves its users.

Web Developer — *Informamuse S.R.L.* *Oct. 2020–Apr. 2021*

Developed websites and web applications using HTML, CSS, JavaScript, PHP, Laravel, Vue.js, and MySQL. Managed projects end-to-end from design through deployment.

ACADEMIC SERVICE

Ethics Reviewer — *NeurIPS 2026* *2026*

Reviewed submissions for ethical concerns, potential societal harms, research-integrity risks, and compliance with the conference's ethics guidelines.

Artifact Evaluation Committee Member — SOSP 2026

2026

Evaluated research artifacts for availability, functionality, reproducibility, and consistency with the experimental claims presented in the associated papers.

Artifact Evaluation Committee Member — MLSys 2026

2026

Reviewed four accepted papers covering attention kernel optimization (ThunderKittens/FlashAttention-3), full-stack ML profiling (XProf), GPU memory disaggregation, and distributed inference. Reproduced experimental results on rented GPU clusters (8×H100 via RunPod) and AMD HPC infrastructure (MI350X through the AMD University Program). Recommended the artifact that ultimately received the Distinguished Artifact Award.

RESEARCH EXPERIENCE

- **DriftBench:** Developed a benchmark measuring how GPU architecture, numerical precision, and inference-framework changes silently alter LLM outputs across 237,000 evaluations. Found that hardware-induced drift is highly predictable ($R^2 = 0.909$), while framework-induced drift generally requires direct re-measurement.
- **GLM TPU Porting:** Ported GLM model workloads to Google Cloud TPUs through the TPU Research Cloud program, addressing model and runtime compatibility issues and validating correct execution on TPU infrastructure. Conducted the work independently as part of a broader investigation into accelerator portability and LLM systems performance.
- **Structured Three-stage Framing:** Conducted approximately 31,900 trials across nine LLMs using a factorial ablation of a professional-framing jailbreak. Found that persona framing primarily determines whether models respond, terminology substitution increases response detail, and moral justification has no positive effect.
- **Contextual Hijacking:** Demonstrated that authority-style formatting in retrieved documents reduces LLM refusal rates for dual-use queries even when content helpfulness is controlled. A pre-specified 2×2 experiment showed a 6.6-percentage-point effect, followed by a blind held-out replication (OR = 0.77); existing classifiers failed to reliably detect the manipulation.

TECHNICAL SKILLS

Languages: Python, JavaScript, PHP, SQL, HTML/CSS, Bash

Frameworks: PyTorch, HuggingFace Transformers

ML/Systems: vLLM, SGLang, TensorRT-LLM, CUDA, Docker

Infrastructure: Linux administration, networking, cloud GPU provisioning (RunPod, AMD HPC/Chameleon Cloud), CI/CD

Research Tools: LaTeX, Git, Jupyter, scipy, statsmodels

COMPUTATIONAL RESOURCES

- Google Cloud TPU Research Cloud (TRC) - active since May 2026. 320 TPU chips: 32 on-demand v4 + 288 spot (v4, v5e, v6e) across US and EU zones.
- AMD University Program account (active since March 2026)
- Chameleon Cloud allocation (2,000 SUs)
- RunPod GPU cluster experience (H100, H200, B200, MI300X configurations)

LANGUAGES

Italian (native), **English** (C1 — academic proficiency, scientific publications)

CONTACTS & PROFILES

Google Scholar: <https://scholar.google.com/citations?hl=en&user=x4K6NWUAAAAJ>

ORCID: <https://orcid.org/0009-0007-6740-2695>

OpenReview: https://openreview.net/profile?id=~Gianluigi_Vitale1

Portfolio: <https://gianluigivitale.github.io/portfolio/>

GitHub: <https://github.com/GianluigiVitale>

LinkedIn: <https://www.linkedin.com/in/gianluigi-vitale615>